

Partial Derivatives and Differentiability Worksheet

1. Consider the surface given by the function $f(x, y) = \sin(xy) + x$.
 - a) What is the equation of the curve C_1 which is the intersection between $f(x, y)$ and $x = 2$?

 - b) Find the slope of the tangent line to the curve C_1 at the point $(2, 0, 2)$.

 - c) What is the equation of the curve C_2 which is the intersection between $f(x, y)$ and $y = 0$?

 - d) Find the slope of the tangent line to the curve C_2 at the point $(2, 0, 2)$.

2. Determine the signs of the partial derivatives for the function f whose graph is shown below.

a) $f_x(1,2)$

b) $f_y(1,2)$

c) $f_x(-1,2)$

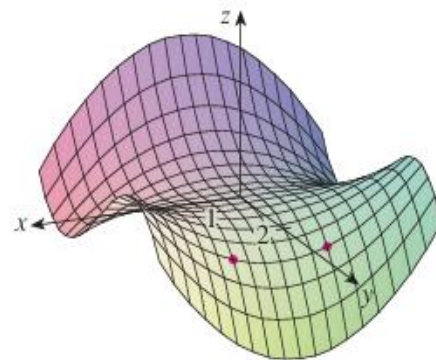
d) $f_y(-1,2)$

e) $f_{xx}(-1,2)$

f) $f_{yy}(-1,2)$

g) $f_{xy}(1,2)$

h) $f_{yx}(-1,2)$



3. Find all the first partial derivatives of the function.

a) $f(x, y) = y^5 - 3xy$

b) $f(x, y) = \sqrt{x} \ln(y)$

c) $f(x, y) = \tan(xy)$

d) $f(x, y) = \frac{x}{(x+y)^2}$

e) $f(x, y) = x \cos(y-x)$

f) $w = \ln(x + 2y + 3z)$

g) $w = ze^{xyz}$

h) $F(x, y) = \int_y^x \cos(e^t) dt$ (Hint: Use the Fundamental Theorem of Calculus).

4. Find all the second partial derivatives of $f(x, y) = x^3y^5 + 2x^4y$.

5. Find all second partial derivatives of $f(x, y) = \sin^2(2x + 3y)$.

6. Find $\frac{\partial z}{\partial x}$ given that $e^{xyz} = 2$.

7. Find the value of $\frac{\partial z}{\partial x}$ at the point $(-1, 2, 1)$ if the equation $xy - z^3x + 5yx = 9$ defines z as a function of the two independent variables x and y and the partial derivative exists.

8. Suppose that $z = f(x, y)$ is determined implicitly by the equation $x^2 - y^2 + z^2 + 2x \sin(y - xz) = 3$. Find $\frac{\partial z}{\partial y}$ and its value at the point $(0, 1, 2)$.

9. Consider the equation $yz + x \ln(y) = z^2$. Find $\frac{\partial z}{\partial x}$.

10. Suppose that for the function $f(x, y)$ we know that f_{xy} and f_{yx} are both differentiable at $(x, y) = (2, 5)$.
Prove that $f_{xy}(2, 5) = f_{yx}(2, 5)$.